

A Criteria-Based Climate Classification Approach Considering Clustering and Building Thermal Performance: Case of India

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Purpose of this document

This is an expanded literature-survey summary prepared for the project's mini-hackathon/review. It summarizes the published research rather than reproducing the copyrighted full paper.

1. Background and Problem

Climate classification is important for building design because climate conditions influence energy efficiency, thermal comfort and the way buildings should be designed. The paper notes that India has an established climate classification framework, but the availability of more detailed and granular weather information creates an opportunity to refine that classification. Traditional approaches may rely heavily on annual averages, which can hide daily and seasonal variations.

2. Aim of the Study

The study develops a criterion-based approach for classifying the climate of locations in India using clustering. A central idea is to use daily climatic variation rather than relying only on annual values. The researchers also validate the resulting climate zones using building thermal performance, linking the statistical grouping to practical building-energy behavior.

3. Data and Variables

The research uses climatic information representing daily conditions and considers variables such as air temperature, relative humidity, solar radiation, precipitation and wind speed. The paper emphasizes that using multiple climatic variables allows clustering to represent climate more comprehensively than approaches based mainly on temperature.

4. Methodology

The study uses K-means cluster analysis. Locations/days with similar climatic characteristics are grouped into clusters. Different values of k are tested rather than assuming one fixed number from the beginning. The resulting classifications are assessed using the silhouette index. The study also considers seasonality, including summer and winter behavior, and uses building thermal-load results as an additional validation perspective.

5. Important Results

The paper reports that the existing five-zone framework was tested first. The proposed clustering-based classification performed better according to the silhouette index: the reported value was 0.21 for the new five-zone classification compared with -0.2 for the existing classification. When different values of k were explored, $k=6$ produced the highest reported silhouette index of 0.3. The resulting classification identified six distinct climate types in the study's framework: hot-dry, hot-humid, warm-dry, warm-humid with cold winters, warm-dry with cold winters, temperate, and cold.

6. Significance

The important contribution is not simply using K-means. The study combines clustering with daily climatic information and practical building-performance validation. It argues that a climate zone should represent meaningful differences in climate behavior rather than being based only on broad annual averages.

7. Limitations / Future Direction

The paper identifies opportunities for further work, including more detailed parametric analysis of climatic factors and additional climatic parameters. The authors also point to the value of continued validation of whether cities grouped in the same climate zone show similar building thermal performance.

8. Connection to Our Project

This paper is highly relevant to our project because it demonstrates K-means clustering on Indian climate data. Our project similarly aims to discover groups of similar climate conditions. The major difference is that our project is designed as a general environmental-data clustering study using an Indian climate dataset with variables such as temperature, humidity, rainfall, wind speed and pressure, and we plan to compare K-means with Hierarchical Clustering. This paper gives us a strong justification for using clustering and for evaluating cluster quality with the silhouette score.

9. Viva Points

If asked what we learned from this paper, we can say: climate classification can be treated as an unsupervised clustering problem; daily/seasonal variation can be important; K-means can reveal meaningful climate groups; and cluster quality should be evaluated rather than choosing k arbitrarily. We should not claim that our project reproduces this paper because our dataset and objectives are different.

Reference: See the publisher record/DOI for the original article.